

ALL TABLES

12 tables · V2 optimal-stopping RMSE summaries

How to read this file

- **What is here:** only the big RMSE summary tables (indexes + compare Quick RMSE). No per-study 1/16 metric blocks, no 3-row rolling mini-tables.
- **Order:** (1) ticker RMSE indexes for SPY → AAPL → MSFT; (2) compare notebooks by rolling mode `none` → `monthly` → `daily`, each with SPY → AAPL → MSFT.
- **Naming:** `TICKER - RMSE index...` = one ticker, all regimes/models, three rolling columns. `compare_rolling_MODE - TICKER - Quick RMSE table` = one rolling mode + one ticker.
- **Rows:** always regime order 2008–2009 → 2013–2014 → 2018–2019 → 2019–2020; within each regime models GBM → Merton → Heston–Merton → GARCH–Merton (16 rows).
- **Index columns:** `Regime`, `Model`, RMSE for `none` / `monthly` / `daily`, `best rolling` (lowest RMSE), `file` (study markdown name).
- **Quick RMSE columns:** `#`, `Regime`, `Model`, RMSE, MAE, Bias (model – market). Same contract sample per ticker (n=24, seed=42); lookback 6 months.
- **Numbers:** lower RMSE/MAE is better. Bold RMSE in compare tables is the headline metric used for ranking.

SPY — RMSE index (none / monthly / daily)

Regime	Model	none	monthly	daily	best rolling	file
2008-2009	GBM	0.6200	0.8139	0.6678	none	2008-2009_gbm.md
2008-2009	Merton	0.6200	0.8320	0.6560	none	2008-2009_merton.md
2008-2009	Heston–Merton	0.5686	0.8097	0.6731	none	2008-2009_heston_merton.md
2008-2009	GARCH–Merton	0.6538	0.6302	0.6990	monthly	2008-2009_garch_merton.md
2013-2014	GBM	0.7724	0.4572	0.4021	daily	2013-2014_gbm.md
2013-2014	Merton	0.7667	0.4786	0.4155	daily	2013-2014_merton.md
2013-2014	Heston–Merton	2.5391	0.6535	0.5309	daily	2013-2014_heston_merton.md
2013-2014	GARCH–Merton	1.0072	0.6103	0.5655	daily	2013-2014_garch_merton.md
2018-2019	GBM	0.9143	1.6168	1.5356	none	2018-2019_gbm.md
2018-2019	Merton	0.8772	1.5545	1.4499	none	2018-2019_merton.md
2018-2019	Heston–Merton	0.8801	1.3155	2.3372	none	2018-2019_heston_merton.md

Regime	Model	none	monthly	daily	best rolling	file
2018-2019	GARCH–Merton	0.7625	3.0739	2.0580	none	2018-2019_garch_merton.md
2019-2020	GBM	2.0618	1.7747	1.7976	monthly	2019-2020_gbm.md
2019-2020	Merton	2.1733	1.8482	1.8588	monthly	2019-2020_merton.md
2019-2020	Heston–Merton	1.4705	1.0868	1.0636	daily	2019-2020_heston_merton.md
2019-2020	GARCH–Merton	7.2019	3.2224	2.6584	daily	2019-2020_garch_merton.md

AAPL — RMSE index (none / monthly / daily)

Regime	Model	none	monthly	daily	best rolling	file
2008-2009	GBM	2.6818	2.2774	2.1442	daily	2008-2009_gbm.md
2008-2009	Merton	2.6862	2.2953	2.0929	daily	2008-2009_merton.md
2008-2009	Heston–Merton	2.1383	4.8871	4.9126	none	2008-2009_heston_merton.md
2008-2009	GARCH–Merton	3.0728	3.3749	2.9586	daily	2008-2009_garch_merton.md
2013-2014	GBM	5.0578	3.5486	3.4746	daily	2013-2014_gbm.md
2013-2014	Merton	4.3754	2.8581	2.9473	monthly	2013-2014_merton.md
2013-2014	Heston–Merton	3.0546	7.5249	7.6005	none	2013-2014_heston_merton.md
2013-2014	GARCH–Merton	12.2803	7.4494	7.2481	daily	2013-2014_garch_merton.md
2018-2019	GBM	0.9192	2.0397	1.8064	none	2018-2019_gbm.md
2018-2019	Merton	0.8663	1.9220	1.7155	none	2018-2019_merton.md
2018-2019	Heston–Merton	0.8859	3.9565	4.8659	none	2018-2019_heston_merton.md

Regime	Model	none	monthly	daily	best rolling	file
2018-2019	GARCH–Merton	0.6696	4.4022	3.9754	none	2018-2019_garch_merton.md
2019-2020	GBM	3.2126	2.1914	2.0972	daily	2019-2020_gbm.md
2019-2020	Merton	3.1599	2.2049	2.0578	daily	2019-2020_merton.md
2019-2020	Heston–Merton	2.2791	6.6880	6.5196	none	2019-2020_heston_merton.md
2019-2020	GARCH–Merton	8.0466	4.1918	4.1158	daily	2019-2020_garch_merton.md

MSFT — RMSE index (none / monthly / daily)

Regime	Model	none	monthly	daily	best rolling	file
2008-2009	GBM	0.3358	0.5822	0.5076	none	2008-2009_gbm.md
2008-2009	Merton	0.3426	0.5647	0.4913	none	2008-2009_merton.md
2008-2009	Heston–Merton	0.3301	1.4610	1.6848	none	2008-2009_heston_merton.md
2008-2009	GARCH–Merton	0.4026	0.9186	1.0189	none	2008-2009_garch_merton.md
2013-2014	GBM	0.2040	0.2378	0.2282	none	2013-2014_gbm.md
2013-2014	Merton	0.2040	0.2305	0.2191	none	2013-2014_merton.md
2013-2014	Heston–Merton	1.3031	0.5959	0.7551	monthly	2013-2014_heston_merton.md
2013-2014	GARCH–Merton	0.2338	0.4485	0.4488	none	2013-2014_garch_merton.md
2018-2019	GBM	0.6724	1.0894	1.0192	none	2018-2019_gbm.md
2018-2019	Merton	0.7817	1.0932	1.0099	none	2018-2019_merton.md
2018-2019	Heston–Merton	2.5064	1.1402	1.2707	monthly	2018-2019_heston_merton.md

Regime	Model	none	monthly	daily	best rolling	file
2018-2019	GARCH–Merton	0.6083	1.3586	1.3519	none	2018-2019_garch_merton.md
2019-2020	GBM	1.4759	0.9960	0.9498	daily	2019-2020_gbm.md
2019-2020	Merton	1.5104	1.0564	0.9964	daily	2019-2020_merton.md
2019-2020	Heston–Merton	1.5278	1.3027	1.1950	daily	2019-2020_heston_merton.md
2019-2020	GARCH–Merton	4.2725	1.5011	1.2630	daily	2019-2020_garch_merton.md

compare_rolling_none — SPY — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.6200	0.4401	0.1531
2	2008-2009	Merton	0.6200	0.4401	0.1531
3	2008-2009	Heston–Merton	0.5686	0.4244	0.1315
4	2008-2009	GARCH–Merton	0.6538	0.4773	0.2535
5	2013-2014	GBM	0.7724	0.6020	0.6020
6	2013-2014	Merton	0.7667	0.5874	0.5846
7	2013-2014	Heston–Merton	2.5391	2.1449	2.1449
8	2013-2014	GARCH–Merton	1.0072	0.8093	0.8093
9	2018-2019	GBM	0.9143	0.6779	-0.6169
10	2018-2019	Merton	0.8772	0.6495	-0.6022
11	2018-2019	Heston–Merton	0.8801	0.6480	-0.5967
12	2018-2019	GARCH–Merton	0.7625	0.5523	-0.4911
13	2019-2020	GBM	2.0618	1.8741	1.8556
14	2019-2020	Merton	2.1733	1.9500	1.9500
15	2019-2020	Heston–Merton	1.4705	1.0153	-0.7097
16	2019-2020	GARCH–Merton	7.2019	6.7792	6.7792

compare_rolling_none — AAPL — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	2.6818	2.2452	1.7229
2	2008-2009	Merton	2.6862	2.2723	1.7733
3	2008-2009	Heston-Merton	2.1383	1.7406	1.0709
4	2008-2009	GARCH-Merton	3.0728	2.6049	2.2066
5	2013-2014	GBM	5.0578	3.8875	3.6376
6	2013-2014	Merton	4.3754	3.4310	3.1359
7	2013-2014	Heston-Merton	3.0546	2.0743	1.1148
8	2013-2014	GARCH-Merton	12.2803	10.2589	10.2589
9	2018-2019	GBM	0.9192	0.7432	-0.6415
10	2018-2019	Merton	0.8663	0.6983	-0.6017
11	2018-2019	Heston-Merton	0.8859	0.6920	-0.5963
12	2018-2019	GARCH-Merton	0.6696	0.5564	-0.2049
13	2019-2020	GBM	3.2126	2.6738	2.6192
14	2019-2020	Merton	3.1599	2.5943	2.5082
15	2019-2020	Heston-Merton	2.2791	1.7419	1.4092
16	2019-2020	GARCH-Merton	8.0466	7.1471	7.1471

compare_rolling_none — MSFT — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.3358	0.2090	-0.1033
2	2008-2009	Merton	0.3426	0.2275	-0.0974
3	2008-2009	Heston-Merton	0.3301	0.2146	-0.0945
4	2008-2009	GARCH-Merton	0.4026	0.3296	0.1598
5	2013-2014	GBM	0.2040	0.1647	0.1250
6	2013-2014	Merton	0.2040	0.1647	0.1250
7	2013-2014	Heston-Merton	1.3031	1.0510	1.0457
8	2013-2014	GARCH-Merton	0.2338	0.1917	0.1596
9	2018-2019	GBM	0.6724	0.4624	-0.3920
10	2018-2019	Merton	0.7817	0.5655	-0.4468
11	2018-2019	Heston-Merton	2.5064	1.9703	1.9703
12	2018-2019	GARCH-Merton	0.6083	0.5206	0.2160
13	2019-2020	GBM	1.4759	1.2637	1.2527
14	2019-2020	Merton	1.5104	1.2809	1.2723
15	2019-2020	Heston-Merton	1.5278	1.2883	1.2883
16	2019-2020	GARCH-Merton	4.2725	3.8936	3.8936

compare_rolling_monthly — SPY — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.8139	0.6378	0.4067
2	2008-2009	Merton	0.8320	0.6569	0.4366
3	2008-2009	Heston–Merton	0.8097	0.6424	0.4524
4	2008-2009	GARCH–Merton	0.6302	0.4605	0.2650
5	2013-2014	GBM	0.4572	0.3682	0.3536
6	2013-2014	Merton	0.4786	0.3845	0.3642
7	2013-2014	Heston–Merton	0.6535	0.4730	0.4366
8	2013-2014	GARCH–Merton	0.6103	0.4818	0.4801
9	2018-2019	GBM	1.6168	1.2303	1.0662
10	2018-2019	Merton	1.5545	1.2042	1.0642
11	2018-2019	Heston–Merton	1.3155	1.0910	0.6527
12	2018-2019	GARCH–Merton	3.0739	2.1487	2.1487
13	2019-2020	GBM	1.7747	1.4128	1.2189
14	2019-2020	Merton	1.8482	1.4788	1.2675
15	2019-2020	Heston–Merton	1.0868	0.8753	0.0601
16	2019-2020	GARCH–Merton	3.2224	2.3905	2.2648

compare_rolling_monthly — AAPL — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	2.2774	1.9362	1.6032
2	2008-2009	Merton	2.2953	1.8750	1.5917
3	2008-2009	Heston–Merton	4.8871	3.9083	3.5237
4	2008-2009	GARCH–Merton	3.3749	2.4869	2.1342
5	2013-2014	GBM	3.5486	2.6324	2.3981
6	2013-2014	Merton	2.8581	2.2008	1.8925
7	2013-2014	Heston–Merton	7.5249	4.9023	3.4576
8	2013-2014	GARCH–Merton	7.4494	5.5961	5.5681
9	2018-2019	GBM	2.0397	1.4420	0.9101
10	2018-2019	Merton	1.9220	1.3646	0.8748
11	2018-2019	Heston–Merton	3.9565	2.8959	2.5233
12	2018-2019	GARCH–Merton	4.4022	2.7817	2.6328
13	2019-2020	GBM	2.1914	1.7784	1.6347
14	2019-2020	Merton	2.2049	1.8114	1.5987
15	2019-2020	Heston–Merton	6.6880	4.6746	4.5283
16	2019-2020	GARCH–Merton	4.1918	3.5054	3.4553

compare_rolling_monthly — MSFT — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.5822	0.4708	0.3315
2	2008-2009	Merton	0.5647	0.4665	0.3270
3	2008-2009	Heston–Merton	1.4610	1.0351	0.8679
4	2008-2009	GARCH–Merton	0.9186	0.7568	0.7402
5	2013-2014	GBM	0.2378	0.1772	0.1415
6	2013-2014	Merton	0.2305	0.1719	0.1344
7	2013-2014	Heston–Merton	0.5959	0.4974	0.4704
8	2013-2014	GARCH–Merton	0.4485	0.3563	0.3255
9	2018-2019	GBM	1.0894	0.7308	0.3609
10	2018-2019	Merton	1.0932	0.7454	0.3607
11	2018-2019	Heston–Merton	1.1402	0.8694	0.4415
12	2018-2019	GARCH–Merton	1.3586	0.9917	0.7258
13	2019-2020	GBM	0.9960	0.7604	0.5521
14	2019-2020	Merton	1.0564	0.7926	0.5889
15	2019-2020	Heston–Merton	1.3027	1.0520	-0.3737
16	2019-2020	GARCH–Merton	1.5011	1.1000	0.9698

compare_rolling_daily — SPY — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.6678	0.5253	0.3093
2	2008-2009	Merton	0.6560	0.5329	0.3436
3	2008-2009	Heston–Merton	0.6731	0.5142	0.3065
4	2008-2009	GARCH–Merton	0.6990	0.4570	0.3876
5	2013-2014	GBM	0.4021	0.3317	0.3199
6	2013-2014	Merton	0.4155	0.3381	0.3065
7	2013-2014	Heston–Merton	0.5309	0.4135	0.3819
8	2013-2014	GARCH–Merton	0.5655	0.4460	0.4381
9	2018-2019	GBM	1.5356	1.1325	0.9813
10	2018-2019	Merton	1.4499	1.0616	0.9334
11	2018-2019	Heston–Merton	2.3372	1.4655	1.0315
12	2018-2019	GARCH–Merton	2.0580	1.5759	1.5733
13	2019-2020	GBM	1.7976	1.4235	1.2793
14	2019-2020	Merton	1.8588	1.4838	1.3243
15	2019-2020	Heston–Merton	1.0636	0.8733	0.1333
16	2019-2020	GARCH–Merton	2.6584	2.1964	2.1923

compare_rolling_daily — AAPL — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	2.1442	1.8399	1.5191
2	2008-2009	Merton	2.0929	1.7377	1.4776
3	2008-2009	Heston-Merton	4.9126	3.7769	3.4199
4	2008-2009	GARCH-Merton	2.9586	2.2743	1.8845
5	2013-2014	GBM	3.4746	2.5452	2.3226
6	2013-2014	Merton	2.9473	2.1999	1.9768
7	2013-2014	Heston-Merton	7.6005	4.6273	4.2590
8	2013-2014	GARCH-Merton	7.2481	5.1280	5.1141
9	2018-2019	GBM	1.8064	1.2107	0.8674
10	2018-2019	Merton	1.7155	1.1588	0.8179
11	2018-2019	Heston-Merton	4.8659	3.8441	3.6048
12	2018-2019	GARCH-Merton	3.9754	2.9800	2.9246
13	2019-2020	GBM	2.0972	1.6606	1.5730
14	2019-2020	Merton	2.0578	1.6737	1.5688
15	2019-2020	Heston-Merton	6.5196	4.8688	4.8212
16	2019-2020	GARCH-Merton	4.1158	3.5165	3.4759

compare_rolling_daily — MSFT — Quick RMSE table

#	Regime	Model	RMSE	MAE	Bias
1	2008-2009	GBM	0.5076	0.3999	0.3276
2	2008-2009	Merton	0.4913	0.4000	0.3111
3	2008-2009	Heston–Merton	1.6848	1.0986	0.9606
4	2008-2009	GARCH–Merton	1.0189	0.7611	0.7541
5	2013-2014	GBM	0.2282	0.1660	0.1224
6	2013-2014	Merton	0.2191	0.1629	0.1134
7	2013-2014	Heston–Merton	0.7551	0.5297	0.5185
8	2013-2014	GARCH–Merton	0.4488	0.3391	0.3034
9	2018-2019	GBM	1.0192	0.6682	0.3543
10	2018-2019	Merton	1.0099	0.6574	0.3690
11	2018-2019	Heston–Merton	1.2707	0.9792	0.7787
12	2018-2019	GARCH–Merton	1.3519	0.9245	0.8098
13	2019-2020	GBM	0.9498	0.7194	0.4940
14	2019-2020	Merton	0.9964	0.7450	0.5084
15	2019-2020	Heston–Merton	1.1950	0.9280	-0.3363
16	2019-2020	GARCH–Merton	1.2630	0.9537	0.7982